

ANNUAL REPORT
SALTERN HAMBANTHOTA

2024

Annual Operations & Production Report: Saltern Management System

Executive Summary

The past fiscal year has been a period of significant operational advancement for the saltern operations, specifically within the Hambantota salt flats. By integrating digital tracking for inventory and weather-based predictive modeling, we have optimized the brine evaporation cycles and enhanced overall yield quality. This report outlines the production metrics, challenges faced due to seasonal fluctuations, and the strategic improvements implemented to maintain a steady supply of industrial and edible salt.

Production and Yield Analysis

Production levels this year remained consistent with targets, driven primarily by favorable solar radiation and wind speeds during the peak dry months. The crystallization process was monitored closely to ensure the purity of Sodium Chloride (NaCl) met industrial standards. Our data indicates that the primary harvest periods yielded high-grade salt, while secondary cycles provided a steady output of gypsum as a valuable byproduct. The implementation of real-time monitoring allowed for better timing of the "raking" process, reducing manual labor overhead and increasing the efficiency of the salt pans.

Environmental and Weather Impact

As salt production is intrinsically linked to meteorological conditions, the year was characterized by distinct seasonal shifts. The integration of weather forecasting models allowed the management to predict rainfall patterns with greater accuracy, enabling proactive measures to protect the brine from dilution. Despite occasional unseasonal showers, the strategic drainage and reservoir management protocols minimized production downtime. High evaporation rates recorded during the second quarter were the primary drivers for the record-breaking harvest seen in that period.

Inventory and Supply Chain Management

The digital transformation of our stock management has provided unprecedented visibility into our supply chain. From inward raw material tracking to outward distribution to major wholesalers, the system ensured zero discrepancies in stock levels. Improved warehouse management protocols have optimized the storage of iodized edible salt and agricultural grade salt, ensuring that moisture content is kept at a minimum. The transition to a more data-driven approach has reduced stock-out situations and improved our response time to large-scale market demands.

Operational Challenges and Future Outlook

While the year was largely successful, challenges such as rising transportation costs and the need for further automation in the packing units were identified. Moving forward, the focus will be on enhancing the Smart Saltern platform with more granular data analytics and exploring sustainable

energy options for the refinery processes. By continuing to bridge the gap between traditional salt harvesting and modern software engineering solutions, we aim to position our operations as a benchmark for the industry in the coming year.